[18MSCH2H04]

Reg. No. 18MSRCH011

SECOND SEMESTER UNIVERSITY EXAMINATION, MAY - 2019

SPECTROSCOPY - II

Time: 03 Hours

Maximum Marks:70

Instructions: I. Question 1 is compulsory, carries 10 marks.
II. Questions 2 to 7 carry 15 marks each, answer any four.

1. Answer any FIVE. 5×2=10

a. Differentiate anhydride from acid using IR spectroscopy.

b. Define conditions to be met for a molecule to be EPR active.

c. State the spin crossover effect for switchable molecules.

d. What are the units of the molar extinction coefficient and absorbance? What are the limitations of Lambert-Beer law?

e. What are the disadvantages of AFM over SEM?

f. Calculate the band gap of a quantum dot having absorption peak at 300 nm?

2. a. How do you differentiate the following using IR spectroscopy:
i. amine and amides
ii. aldehydes and ketones
iii. carboxylic acids and esters.

b. Explain the IR frequencies of alcohols, phenols, and ethers with examples. 6+6

3. a. Discuss the typical ESR spectra of V^{4+} and Cu^{2+} and discuss the significance of $A_{\parallel}$ and $g_{\parallel}$. 10+5

b. Explain the significance of $A_{\parallel}$ and $g_{\parallel}$ to analyze the metal complex.

4. a. Describe the isomeric shift, quadruple and magnetic splitting in Mössbauer spectroscopy. 10+5

b. Explain the basic principle involved in NQR and list out its applications.

5. a. Explain the basic differences between UV spectrophotometer and spectrofluorimeter? Draw schematically how the detector, sample holder and source light are aligned in the spectrophotometer and spectrofluorimeter, respectively. What is the main physical reason behind such orientations?

b. How one can calculate the Stern Volmer constant from an experimental result? What is the unit of Stern Volmer constant? 10+5

6. a. Illustrate the working principle and instrumentation of atomic-force microscopy (AFM) with a neat diagram.

b. What are the applications of atomic-force microscopy (AFM)? 10+5

7. a. A quantum dot showed an absorbance of 0.5 at 400 nm in a cell of path length of 1 cm under a UV-vis spectrophotometer. Calculate the concentration of the quantum dot? The same solution showed twice emission intensity (I_S/I_R=2) and twice absorbance (A_S/A_R=2) than that of a standard quinine sulphate (having quantum yield of 0.54) in water medium. Calculate the quantum yield of the quantum dot?

$\varepsilon = 50,000$ ~~mol⁻¹cm⁻¹~~ $M^{-1}cm^{-1}$

b. Draw the Jablonski diagram depicting potential wells of the LS and HS sates. 10+5

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